

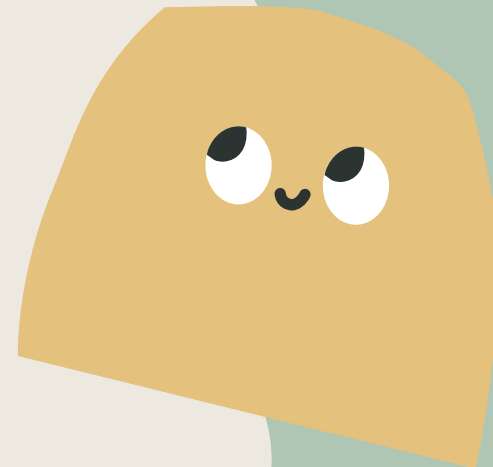
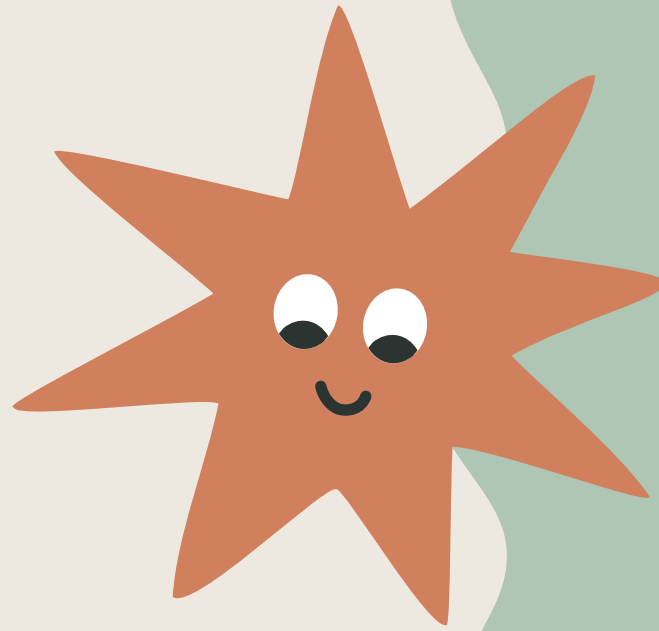
UNIT 1

FUNDAMENTALS OF

COMPUTER

9TH COMPUTER SCIENCE
SINDH TEXTBOOK BOARD

AWINASH GOSWAMI



Agenda

Evolution of Computers

Classification of Computers

Role of Computer

Careers in IT Field

Computer Hardware

Basic Operations of A Computer

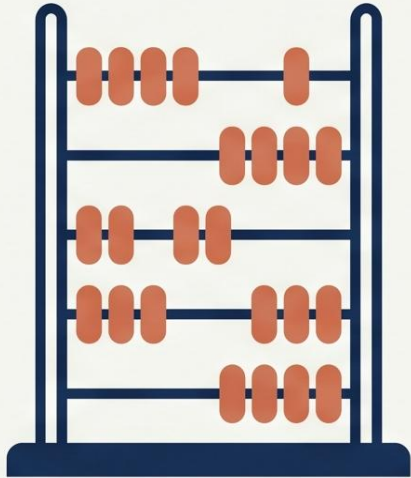
Computer Software

Introduction to computer

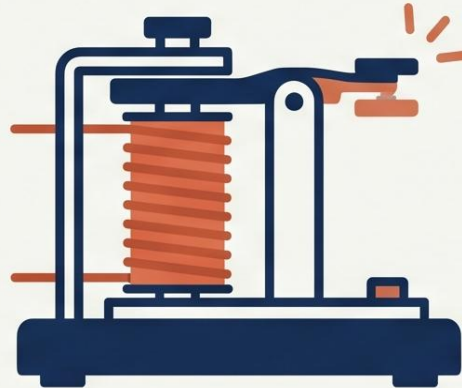
A computer is an electronic data processing machine.

EVOLUTION OF COMPUTERS

MECHANICAL ERA
(DARK AGE)



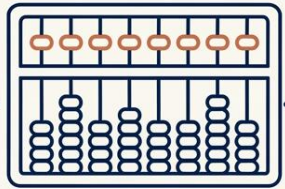
ELECTRO-MECHANICAL
ERA (MIDDLE AGE)



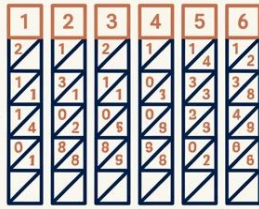
ELECTRONIC ERA
(MODERN AGE)



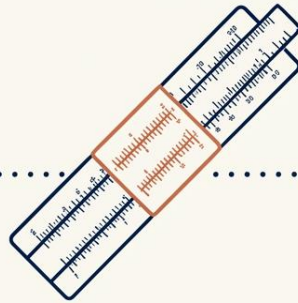
MECHANICAL ERA (DARK AGE)



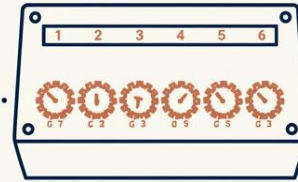
Abacus
(3000 B.C.)



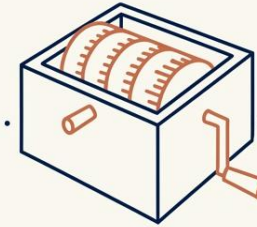
Napier's Bones
(1612 A.D.)



Slide Rule
(1622 A.D.)



Pascaline
(1642 A.D.)



Leibniz's Calculator
(1694 A.D.)



Babbage's Engine
(1822 & 1837 A.D.)

Abacus (3000 B.C.)

- Abacus was invented about 5000 years ago.
- It is also known as counting frame.
- Abacus is still used to teach basic arithmetic operations to the students.
- Abacus is considered as first computer prototype.



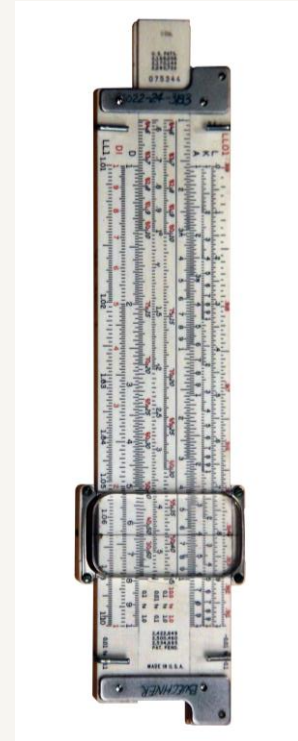
Napier's Bones (1612 A.D.)

- Scottish mathematician John Napier developed Napier's Bones, in 1612.
- It is also called Napier's Rods.
- It was a small machine that contained ten rods. The rods were made up of bones, wood or metal.
- Napier's Bones was used for multiplication and division.



Slide Rule (1622 A.D.)

- English mathematician William Oughtred invented a Slide Rule.
- This was basically a pair of two moveable rules placed side by side.
- They had a standardized set of markings called scales. These scales were used for mathematical computations.
- Slide Rule was used to multiply and divide numbers. The modern slide rules are still used in some countries.



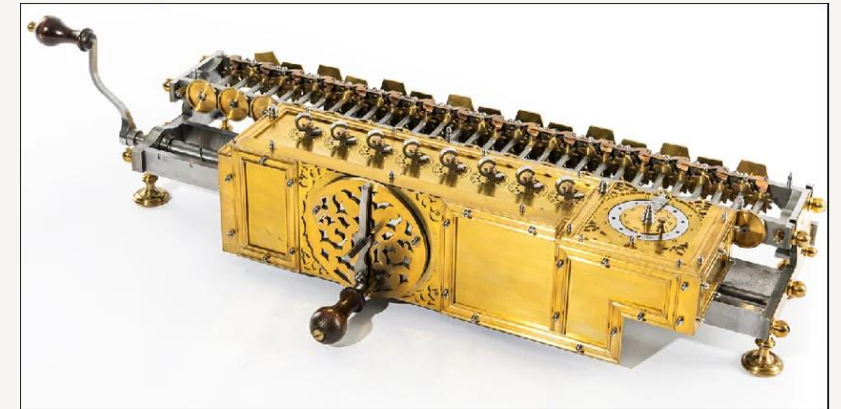
Pascaline or Pascal's Calculator (1642 A.D.)

- French mathematician Blaise Pascal invented a mechanical calculator which was called Pascaline.
- He developed this machine to help his father who was a tax collector.
- Pascaline can perform addition and subtraction by eight figures. User had to adjust wheels in order to perform these calculations.



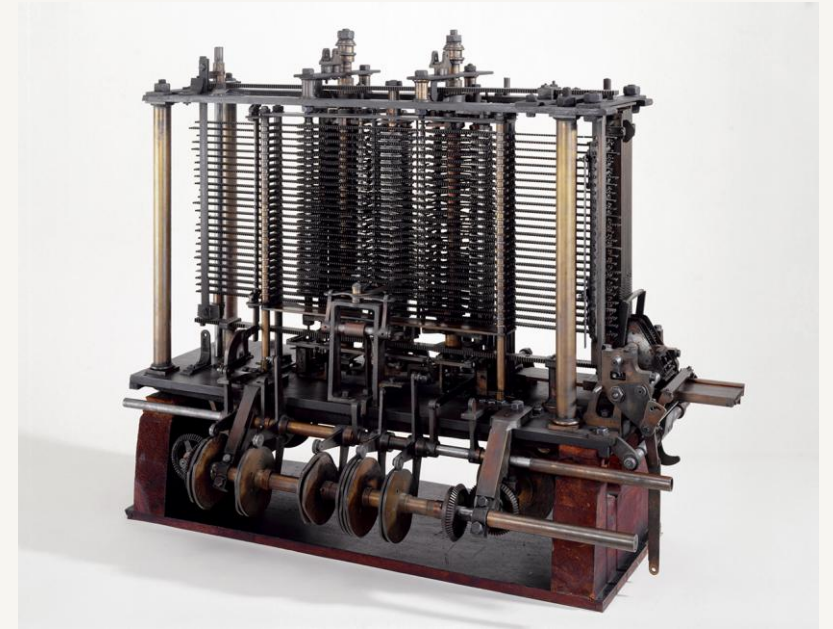
Leibniz's Calculator (1694 A.D.)

- Leibniz's Calculator was developed by the German mathematician Gottfried Wilhelm Leibniz.
- It was also called stepped reckoner.
- It was the first calculator that could perform all four basic arithmetic operations; addition, subtraction, multiplication and division



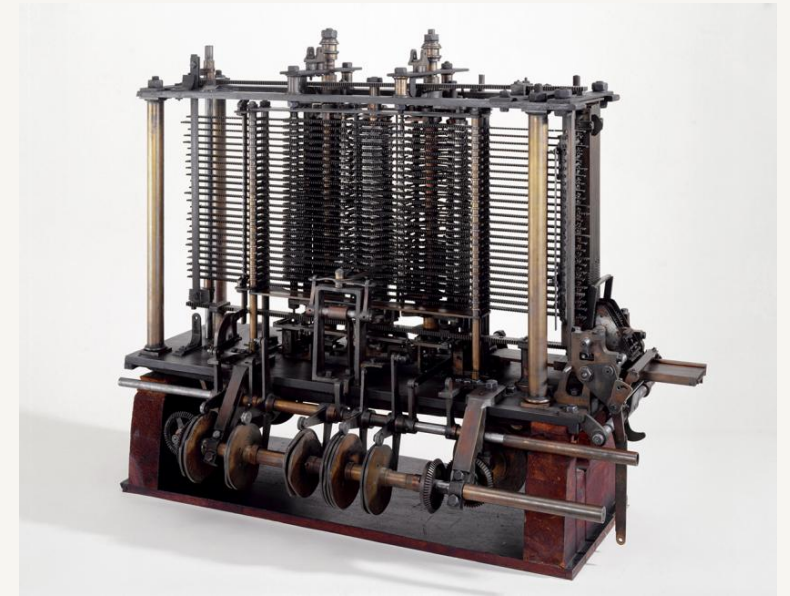
Babbage's Difference and Analytical Engines (1822 and 1837 A.D.)

- Charles Babbage, an English mathematician and engineer, is called the Father of Computers for creating the first complete computing machine.
- His first invention, the Difference Engine (powered by steam), was a large metal calculator that:
 - *Stored data temporarily in mechanical memory.*
 - *Printed results from calculations using difference equations.*

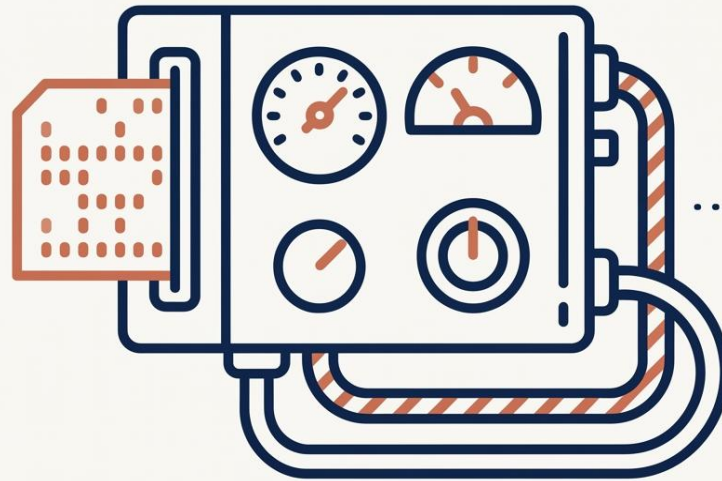


Babbage's Difference Cont....

- In 1837, he designed the Analytical Engine, which he never built. Its advanced design included:
 - *An ALU (Arithmetic Logic Unit).*
 - *Programmatic flow control.*
 - *Punched-card programming and integrated memory.*
- Historians view it as the first concept of a general-purpose computer due to its comprehensive design.



ELECTRO-MECHANICAL ERA (MIDDLE AGE)



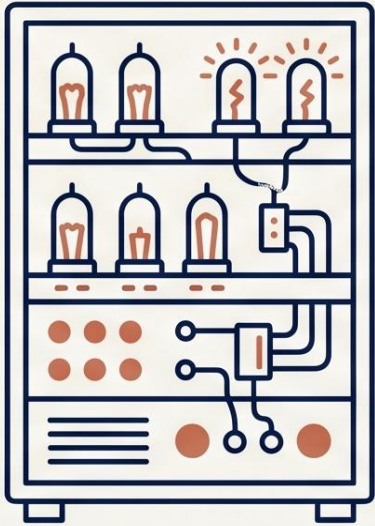
Hollerith's Tabulating Machine (1890 A.D.)

Herman Hollerith's Tabulating Machine (1890 A.D.)

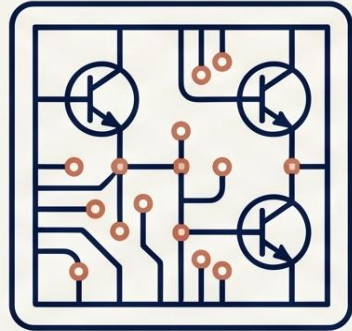
- In 1890, American inventor Herman Hollerith invented the Tabulating Machine. It was first used for the U.S. Census that year.
- While the 1880 census took 7 years to complete, Hollerith's machine finished the 1890 census in just 6 months—a huge speed boost.
- The machine became very popular and was widely used in many U.S. government offices.



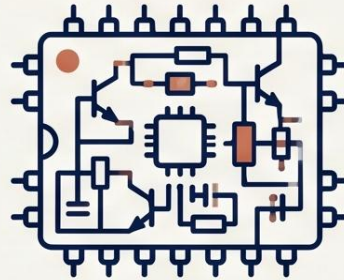
ELECTRONIC ERA (MODERN AGE)



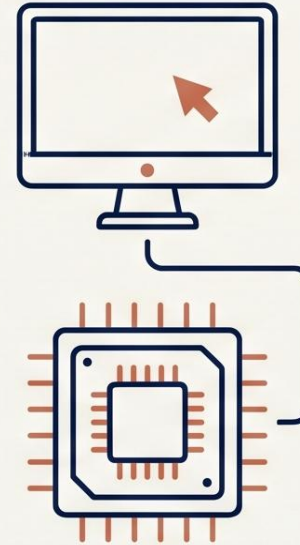
First Generation
(1940–1956)



Second Generation
(1956–1963)



Third Generation
(1964–1971)



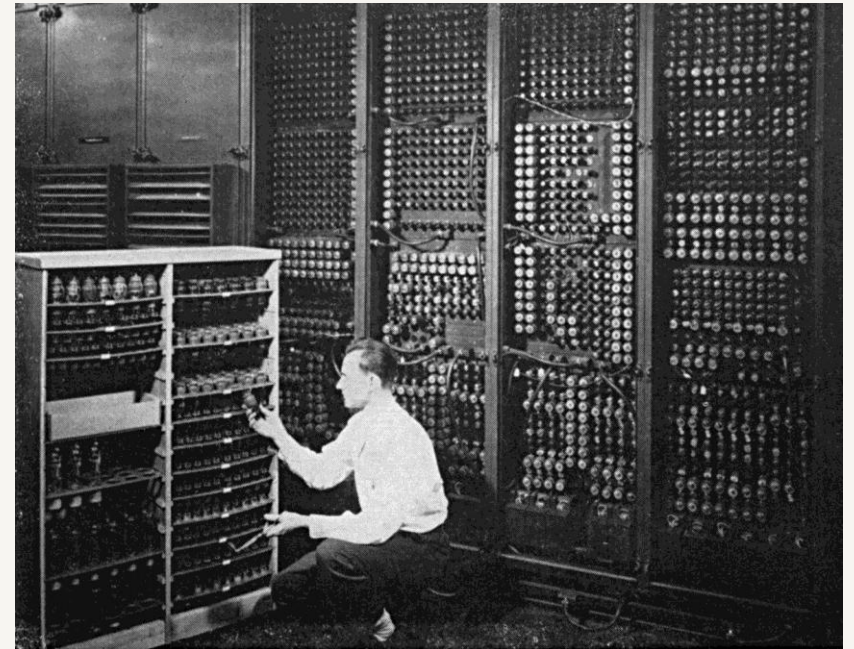
Fourth Generation
(1971–Present)



Fifth Generation
(Present and Beyond)

First Generation of Computers (1940 to 1956)

- First-generation computers (based on vacuum tubes) were very large in size. They used:
 - *Machine language (1s and 0s) for programming.*
 - *Magnetic drums as primary internal storage.*
 - *Punched cards for input.*
 - *Batch processing operating systems.*
- These computers were mainly used for scientific and research purposes. Examples are ENIAC and UNIVAC.



Second Generation of Computers (1956 to 1963)

- Second-generation computers used transistors, making them smaller, faster, cheaper, and more efficient than first-generation machines.
- Key features:
 - *Assembly language and high-level language FORTRAN were introduced.*
 - *Magnetic core replaced drums as primary internal storage.*
 - *Punched cards continued for input.*
 - *Batch processing and multiprogramming operating systems were used.*
- These computers were used for commercial production, scientific, and engineering work. Examples: IBM 7094 and IBM 1401.



Third Generation of Computers (1964 to 1971)

- Third-generation computers used Integrated Circuits (ICs), which further decreased size and increased speed and efficiency while making computers less expensive.
- Key features:
 - *High-level languages like Pascal and COBOL were used.*
 - *Keyboard (input) and monitor (output) made computers easier to use.*
 - *Time-sharing and real-time operating systems were introduced.*
 - *Computers were now used for database management and automatic industrial control.*
- Examples: IBM 360 and IBM 370.



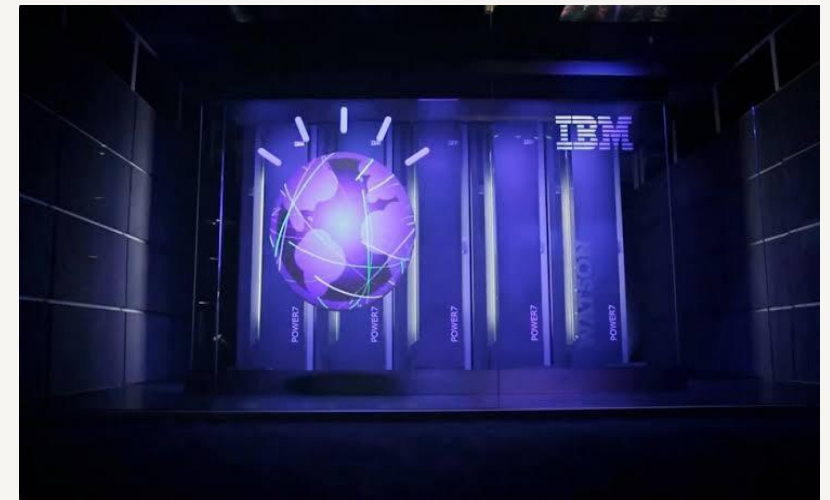
Fourth Generation of Computers (1971 to Present)

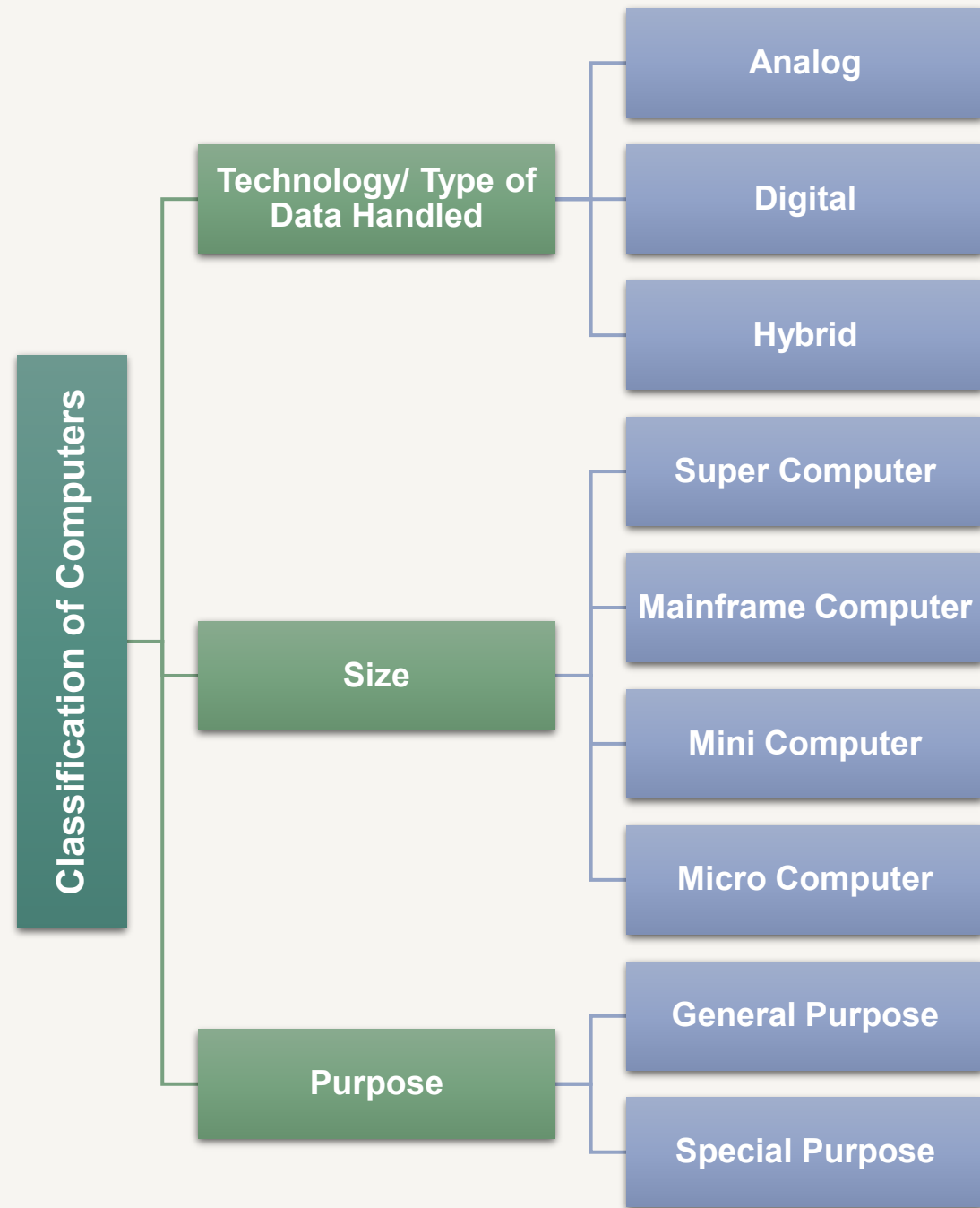
- Fourth-generation computers used microprocessors, which was revolutionary. They became faster, cheaper, smaller, and more reliable.
- Key features:
 - *Used semiconductor memories (RAM and ROM); magnetic storage became popular.*
 - *More high-level languages introduced – C, C++, Java, etc.*
 - *Used in almost every field – space, business, art, and more.*
 - *Operating systems: time-sharing, real-time, and distributed.*
 - *Saw the development of Graphical User Interfaces (GUIs) for easier use.*
- Examples: Apple Macintosh, IBM PC.



Fifth Generation of Computers (Present and Beyond)

- Fifth-generation computers are still being developed. They aim to make computers capable of self-learning, reasoning, and generalization.
- Key features:
 - *Will be able to process human languages and control machines like robots.*
 - *Based on Artificial Intelligence (AI) , which includes branches like Machine Learning, Deep Learning, Natural Language Processing, Robotics, Expert Systems*
- These computers represent the future of intelligent and human-like computing.





TECHNOLOGY BASED



Analog Computers

Analog Computers process analog data — continuously varying physical quantities like pressure, temperature, voltage, speed, and weight.

Examples: Car speedometer, voltmeter.



Digital Computers

Digital Computers are the most commonly used type of computers. They process information using the binary number system (0s and 1s). They are used in homes, educational institutes, offices, businesses, and scientific fields.



Hybrid Computers

Hybrid Computers are a combination of Analog and Digital computers. They combine both features in a single machine and use analog-to-digital and digital-to-analog conversion. They can input or output either digital or analog data. Example: ICU monitors in hospitals and modern petrol pumps.

SIZE BASED

Super Computers

Supercomputer = fastest, largest, most powerful computer — used for complex scientific tasks like weather forecasting and nuclear research.
Example: PARAM (India), Fugaku (Japan).

Mainframe Computers

Mainframe = powerful, multi-user computer for large organizations (like banks) that handles huge calculations at high speed.
Example: IBM zSeries used by banks for transaction processing.

Minicomputer Computers

Minicomputer = mid-sized, multi-user computer — stronger than a PC, smaller than a mainframe. Used as network servers.
Examples: DEC VAX, IBM AS/400.

Microcomputers Computers

Microcomputer = personal computer (PC) for individual use — cheap, fast, and reliable.
Examples: Desktop, laptop, smartphone.

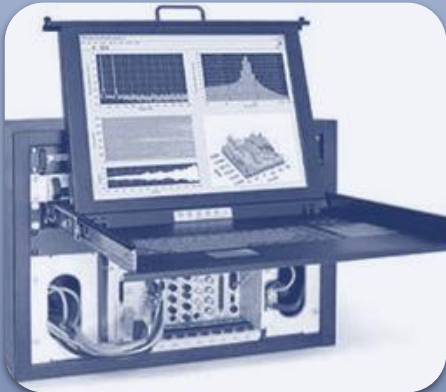
PURPOSE BASED



General Purpose Computers

General Purpose Computers are the most common computers today. They can perform a wide variety of tasks like word processing, payrolls, accounts, inventory control, database management, scientific calculations, statistical analysis, and security control.

Examples: Desktops, laptops, tablets, smartphones.



Special Purpose Computers

Special Purpose Computers are designed to perform specific tasks and repeat a single job more efficiently. They are also called Dedicated Computers. They are useful in traffic light control systems, navigational systems, aviation, weather forecasting, satellite tracking, and ATMs.

ROLE OF COMPUTER



Artificial Intelligence (AI)

AI = machines that **think and act like humans**.
Machine Learning = AI that **learns from data**.
Used in: science, healthcare, space tech.



Robots

Robots = **computer-controlled machines** that do tasks automatically.
Robotics = **making and programming** those robots.
Used in: factories, medicine, science, education.



Wireless Communication

Wireless communication = **sending data through waves** (no wires).
Powers: GPS, smartphones, 3G/4G/5G internet.



Virtual Reality

VR = a **computer-made fake world** that feels real.
Used for: training, education, games, interactive stories.

Careers In IT Field

Career	Role / Description
Software Engineer	Uses programming languages to develop software like games, LMS, business, educational, and entertainment software.
Network Administrator	Manages an organization's network – installs, maintains, and upgrades hardware/software for efficient networking.
Database Administrator (DBA)	Maintains a secure database environment in an office, business, or organization.
Web Designer	Builds websites with attractive designs, images, and text. Responsible for how a site looks, works, and stays updated.
Graphic Designer	Creates layouts for brochures, magazines, ads, and documents using graphics software – combines art and technology.
Information Security Analyst	Protects networks and prevents data loss. Called the "police force of the internet" – stops hackers from stealing account details.
Computer Science / IT Teacher	Teaches computers in schools, colleges, and universities – shows students the importance and impact of computers.

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graph TD; A[Computer Hardware] --> B[System Unit]; A --> C[Input Devices]; A --> D[Output Devices]
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Computer
Hardware

System
Unit

Input
Devices

Output
Devices

Computer Hardware: System Unit

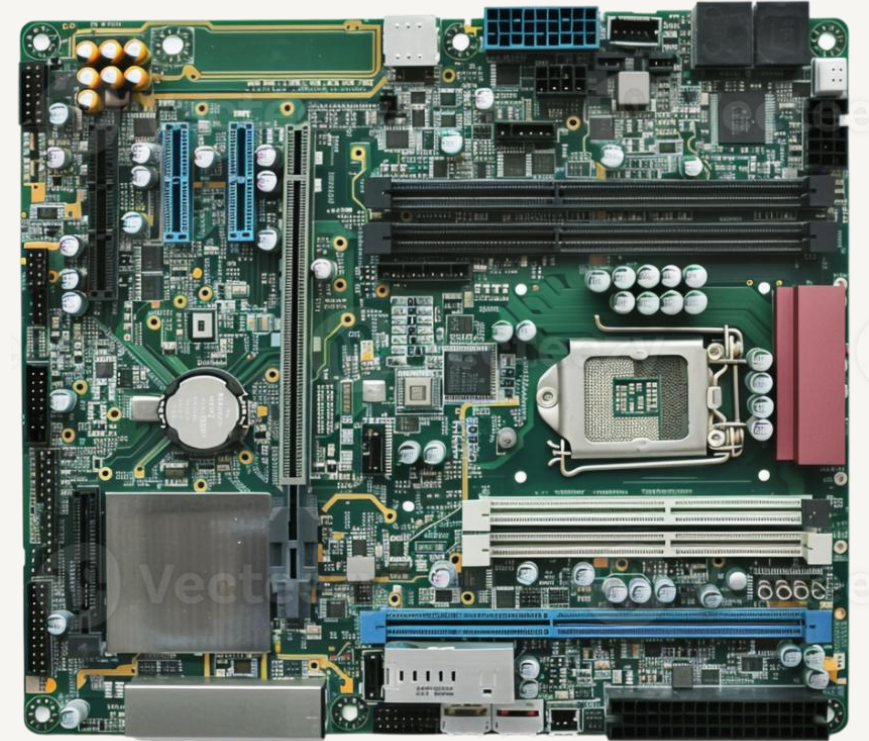
- A System Unit is the part of a computer that contains the primary devices. The system unit includes the motherboard, CPU, RAM and other components, as well as the case in which these devices are enclosed.



System Unit: Motherboard

- Motherboard = **main circuit board** connecting all computer parts (CPU, RAM, slots, ports). Like the **nervous system** — lets everything communicate.

Example: The board inside your PC where the processor, RAM, and graphics card are plugged in.



System Unit: Microprocessor (CPU)

- The CPU (microprocessor) is the brain of the computer. It is a chip containing millions of tiny transistors made of silicon that manipulate data. It performs all calculations at very high speed and accuracy, and fetches, decodes, executes, and stores instructions. The computer's speed mainly depends on the microprocessor's speed — faster clock speed (GHz), more cores, and more cache make it faster.
- Five main components: ALU, Control Unit, Clock, Registers, and Cache.



Arithmetic Logic Unit (ALU)

ALU performs all the actual calculations like arithmetic operations and logical comparisons. Arithmetic operations include addition, subtraction, multiplication and division while logical comparisons include comparing, selecting and matching of data.

Control Unit (CU)

Control Unit is responsible for controlling the transfer of data and instructions among other units of a computer. This unit controls the operations of all parts of the computer but does not carry out any actual data processing operations. CU functions just like a traffic policeman. It manages and coordinates all the units of the computer.

Clock

Clock generates pulses and instructions are executed on the basis of pulses. Clock speed is measure in MHz and GHz

Registers

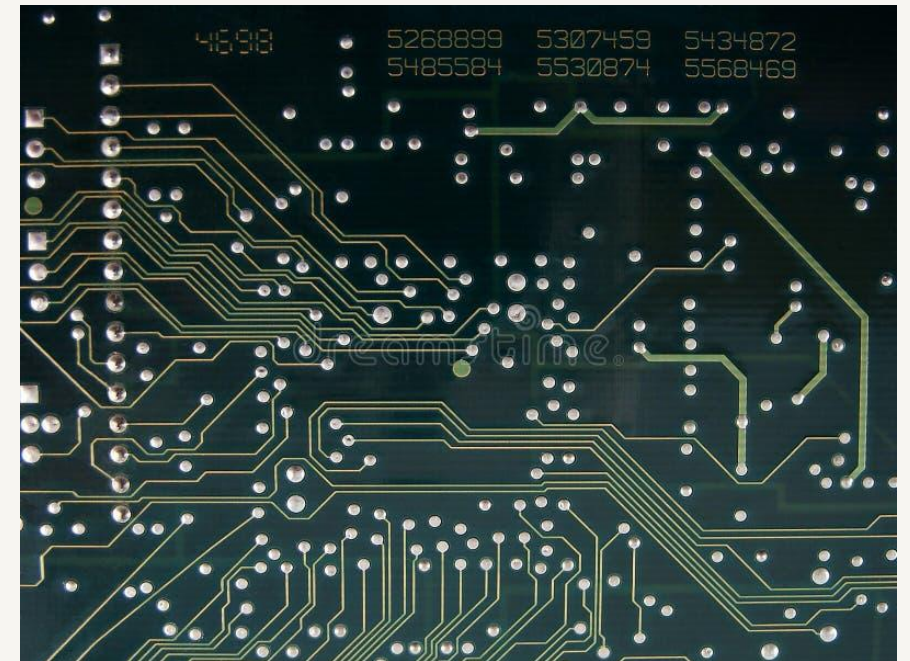
It is a temporary storage area that holds the data that is being processed. It is also known as programming model which may be of 8 bits, 16 bits, 32 bits or 64 bits.

Cache

Cache is an intermediate storage area, which is available inside microprocessor. The immediate processed information is stored in cache. The cache inside the microprocessor is called internal cache and outside is called external cache.

System Unit: Buses

- **Buses** are **electrical paths** on which data is sent and received by different computer components. They are like **roads** — just as roads connect places, buses connect all parts of the computer and internal components on the motherboard.
- **Three types:** Control bus, Data bus, and Address bus.



Control Bus

carries command between different components to control all activities in a computer.

Data Bus

carries data between the processor, memory unit and other components.

Address Bus

carries the address of the data (but not the data). The address bus is used to specify memory location to be used by microprocess for specific operation.

Computer Hardware: Input Devices

- Input devices are used to enter data into computers. These devices can be categorized into Text Entering, Pointing and Image Scanning Devices.

Keyboard

Keyboard = **most common input device** for typing text. Each key press sends a **scan code** to the CPU.

Example: The keyboard you use to type documents or browse shortcuts.

Pointing Devices (Mouse and its variants)

They are used for the quick movement of cursor on screen needed usually in graphic mode. It includes mouse, joy stick, track ball and track pad.

Image Scanners

Scanner = **converts printed images into digital format** using light.
Types: OMR, OBR, OCR.

Examples: Barcode scanners at stores, OCR scanning printed text into editable documents.

Other Types of Input devices

Other input devices = **microphone** (sound/voice), **touch screen** (touch input), **MICR & magnetic strip reader** (specialized input).
Examples: Voice typing on phones, ATMs using touch screens, banks using MICR on cheques.

Computer Hardware: Output Devices

- A hardware device that sends data from a computer (CPU) to another device or user is called an output device. The most common output devices of a computer are monitor and printer. There are two types of output devices.

BASIC OPERATIONS OF A COMPUTER

Input

Computer input has many forms. It can be from a command entered using a keyboard or a mouse.

Processing

Processing is the conversion of input into output.

Storage

Storage refers to the holding or saving data. RAM, ROM and Hard Disk are the devices that are used for storage.

Output

Output is the result of a computer after processing. Output may be viewed on a monitor screen, heard through speakers or printed on paper.

COMPUTER SOFTWARE

- Software is a set of instructions that a computer uses to perform a task. It is a general term for a computer program. There are two main types of software; System Software and Application Software.

COMPUTER SOFTWARE: System Software

- **System Software** is a computer program that **coordinates all activities and functions** of a computer and **controls hardware operations**. It includes:

Operating Systems

The master control program that manages all system resources and creates an interface between the user and the computer.
Examples: Windows, Linux.

Device Drivers

Programs that control a particular device connected to a computer. They act as a translator between the OS and hardware.

COMPUTER SOFTWARE: System Software Cont...

Utility Programs

They are useful programs that help manage, maintain, and control computer resources. Operating systems have some pre-installed, but utility software offers more functionality.

Examples:

- Antivirus – protects from viruses and harmful files.
- Hard Disk Tools – manage storage devices, scan for problems.
- Disk Cleaner – removes unnecessary files.
- Disk Defragmenter – reorganizes files to improve disk performance.

Language Translators

They convert human-readable instructions into machine language (0s and 1s). Software is usually written in high-level languages, and translators convert it for the computer.

Three types: Assembler, Compiler, Interpreter.

COMPUTER SOFTWARE: System Software – Cont...

Assembler

The assembler translates the program written in assembly language into machine language instructions for execution.

Compiler

It translates the entire high-level language program at once into machine language before it is executed.

Interpreter

It translates the high-level language program line by line into machine language.

COMPUTER SOFTWARE: Application Software

- Application Software is used to complete specific tasks like creating documents, databases, spreadsheets, and presentations. Computer games, media players, and web browsers are also examples.
- Must be installed separately — not pre-installed on the OS.
- *Example:* Windows alone can't make spreadsheets — you need MS Excel.
- There are different types of application software.

COMPUTER SOFTWARE: Application Software

Productivity Software

This kind of application software is used to produce things such as documents, spreadsheets, databases and presentations. Many of the productivity software are intended to be used in businesses and offices. MS Office for Windows is an example of productivity software.

Business Software

Business Software are used to manage business activities efficiently. It is used to provide many business functions such as billing, database management and inventory management.

Entertainment Software

This type of software provides amusement and fulfills a user's hobby. The most common entertainment software are video games.

Educational Software

Educational software is used for teaching and learning. This kind of software is used in many schools to enhance knowledge of students in learning different subjects.